

Beyond Formality: The Firm-Size Wage Premium in Colombia

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Abstract

Firm-size wage premia are well documented in advanced economies, but their magnitude and interpretation in middle-income labor markets with large informal sectors remain less settled. This paper documents the firm-size wage premium in Colombia using harmonized microdata from the Gran Encuesta Integrada de Hogares (GEIH) covering 2008 to 2025, excluding 2020. Employment is concentrated in solo work and micro-firms, while real hourly wages rise sharply with firm size. After controlling for gender, age, age squared, education, formality, and sector-year fixed effects, workers in firms with 101 or more employees earn about 43 percent more than otherwise comparable solo workers. The premium is positive throughout the sample period, with no evidence of secular decline. Because the premium survives formality controls, it is not simply a relabeling of the formal-informal wage gap. More specifically, there is a steep size gradient within informal employment, and a narrower large-firm advantage within formal employment relative to intermediate-size firms. Combined with the thin middle of the Colombian firm-size distribution, the premium is consistent with labor-market frictions that may constrain the growth of small businesses and limit worker reallocation toward larger, potentially more productive firms.

1 Introduction

Firm size is a well-established correlate of wages in labor economics (Brown and Medoff, 1989; Oi and Idson, 1999; Troske, 1999). In labor markets where many workers are own-account workers or employed in micro-firms, wage differences between small and large firms help identify where questions about firm growth, worker allocation, and wage inequality are most relevant. This paper asks whether Colombian workers in larger firms earn higher real hourly wages than observably similar workers in smaller firms, how much of that gap is explained by observable worker characteristics, whether the gap has increased or decreased over time, and whether the wage gap between small and large firms differs between formal and informal employment.

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Our motivation to measure the firm-size wage premium in Colombia is both empirical and policy-oriented. A recent policy diagnosis argues that low-income households are disproportionately linked to own-account work, very small businesses, informality, and low-earning jobs, whereas higher-income households are more often connected to larger and more formal employers (Eslava and Melendez, 2026). This diagnosis is suggestive, but it does not show whether firm size remains associated with higher wages after accounting for worker characteristics and formality. The observed link between larger employers and higher household income may instead reflect the fact that larger firms employ workers with different characteristics, or that larger firms are more likely to offer formal jobs. This distinction matters because the wage gap associated with firm size has not been systematically documented in Colombia. The question is especially relevant in Colombia because employment remains concentrated outside the largest-firm categories, while recent growth theory links the rise of large firms to economic development (Chen, 2026).

We study the firm-size wage premium in Colombia using harmonized microdata from the Gran Encuesta Integrada de Hogares (GEIH) for 2008–2019 and 2021–2025. We construct a comparable firm-size classification, measure real hourly labor income, document the distribution of employment across firm sizes, and estimate wage gaps that progressively condition on sector-year effects, worker characteristics, and formality. Because unobserved worker characteristics may be correlated with firm size, the estimates are conditional wage gaps rather than causal effects of firm size. This interpretation follows the broader firm-size wage premium literature: a premium that survives observable controls is informative about sorting, segmentation, firm-side pay differences, and possible allocative frictions, but it is not by itself proof of any one mechanism (Brown and Medoff, 1989; Oi and Idson, 1999; Abowd et al., 1999; Card et al., 2013; El Badaoui et al., 2010; Balkan and Tumen, 2016).

The evidence delivers five main results. First, employment is concentrated at the bottom of the firm-size distribution: solo workers and firms with up to ten workers account for about two thirds of employment. Second, raw wage gaps are large across firm-size categories, sectors, education groups, and formality status, while the wage gap by sex is small by comparison. Third, the firm-size wage premium attenuates substantially after adding worker controls, sector-year fixed effects, and formality, but it does not disappear: in the full specification, workers in firms with 101 or more employees earn about 43 percent more than otherwise comparable solo workers. Fourth, the conditional premium is positive throughout the period; the point estimates suggest an increase between 2008 and 2025, but the change is not statistically significant. Fifth, formality status affects the firm-size wage premium. Formal workers earn more than informal workers within every firm-size category, but the conditional firm-size wage premium is much steeper among informal workers. Within formal employment, the conditional premium relative to formal solo workers is small, although workers in 101+ firms earn more than workers in intermediate-size formal firms.

The rest of the paper is organized as follows. Section 2 describes the GEIH data, the harmonization of firm-size categories, and the construction of real hourly labor income. Section 3 presents the descriptive evidence on employment shares and wage gaps by firm size, formality, education, gender, sector, and year. Section 4 introduces the econometric specifications. Section 5 reports the main regression results, the dynamic estimates, the formality interactions, and the elasticity benchmark, contrasting each result with the relevant literature. Section 6 discusses interpretation, limitations, and policy implications. Section 7

concludes.

2 Data

The empirical analysis uses microdata from the Gran Encuesta Integrada de Hogares (GEIH), Colombia’s nationally representative labor force survey produced by DANE. The current harmonized file covers the years 2008–2019 and 2021–2025. We exclude 2020 because the COVID-19 pandemic disrupted both labor-market outcomes and survey collection, raising comparability and data-quality concerns. The analysis therefore focuses on non-pandemic survey years. The main qualitative patterns documented below are visible on both sides of the excluded year: the firm-size wage premium is positive in the pre-pandemic period and remains positive after 2020. The working sample contains 5,237,255 worker-year observations with valid real hourly labor income and positive expansion weights. The baseline regression sample contains 5,154,213 observations after requiring non-missing values for the controls used in the econometric specification.

The sample is restricted to employed workers with positive labor income, valid weekly hours, and positive survey expansion factors. Monthly labor income is converted to annual labor income and divided by annual hours, computed as weekly hours times 52. We then express hourly labor income in constant 2025 Colombian pesos using the end-of-the-year Consumer Price Index:

$$w_{it} = \frac{12 \times MonthlyIncome_{it}}{52 \times WeeklyHours_{it}} \times \frac{CPI_{2025}}{CPI_t}. \quad (1)$$

Firm size is measured using the worker-reported firm-size variable in GEIH. The construction harmonizes the categories into the following ordered groups: Solo, 2-3, 4-5, 6-10, 11-19, 20-30, 31-50, 51-100, and 101+. When the newer questionnaire separates 101-200 and 201+ workers, both categories are collapsed into 101+ to maintain comparability with earlier years. Firm size is therefore reported by workers, and the data do not identify firms.

The control variables are also harmonized across survey changes. Education is grouped into comparable categories: no education, preschool, primary, lower secondary, upper secondary, and higher education. Formality is defined using pension contribution status: workers who contribute to pensions are classified as formal, while workers who do not contribute are classified as informal. This definition follows the social-protection approach commonly used in Latin American labor-market measurement, which identifies informality through lack of access to social-security benefits such as pension or health insurance (World Bank, 2026). Economic activity is harmonized across CIU revisions into broad sectors, which allows the regressions to absorb sector-year shocks.

3 Descriptive Statistics

The descriptive section is organized around the empirical facts that motivate the regression analysis. We first document where workers are located in the firm-size distribution and how wages per hour vary across that distribution. We then place the raw firm-size wage gap next to other remuneration gaps, describe worker composition by firm size, and examine

whether the raw firm-size pattern appears within formality status, sex, education groups, and economic sectors.

Table 1 establishes the two core raw facts behind the paper. First, mean real hourly income rises sharply with firm size. In 2008, solo workers earned 5,388 COL 2025 pesos per hour on average, compared with 12,943 pesos among workers in firms with 101 or more employees, a ratio of 2.4 to 1. By 2025, the corresponding means were 6,534 and 16,859 pesos, a ratio of 2.6 to 1. The all-worker mean increased from 7,752 to 10,295 pesos, equivalent to 1.7 percent annualized real growth between 2008 and 2025. Average weekly hours in the analytic sample declined from 47.8 to 43.8 over the same period, so real hourly wages grew faster than implied labor income per worker, which increased by 1.5 percent per year. Second, employment remains concentrated at the bottom of the firm-size distribution. In 2025, solo workers accounted for 35 percent of employment, and workers in firms with up to ten workers accounted for 61 percent. The middle of the distribution was thin: firms with 11–100 workers jointly accounted for 14 percent of employment. The 101+ category grew from 19 percent of employment in 2008 to 25 percent in 2025, but this increase did not eliminate the large employment mass in solo work and micro-firms.

Table 1: Employment shares and mean real hourly income, 2008 vs. 2025

Firm size	Employment share		Mean hourly income		Wage ratio to solo		Wage growth
	2008	2025	2008	2025	2008	2025	Annualized
Solo	36.1%	34.7%	5,388	6,534	1.0	1.0	1.1%
2-3	19.6%	13.8%	6,242	7,242	1.2	1.1	0.9%
4-5	7.3%	7.0%	7,196	8,242	1.3	1.3	0.8%
6-10	5.9%	5.8%	7,543	9,663	1.4	1.5	1.5%
11-19	3.4%	3.9%	8,679	11,634	1.6	1.8	1.7%
20-30	3.3%	3.3%	8,905	11,201	1.7	1.7	1.4%
31-50	2.6%	3.0%	10,424	12,574	1.9	1.9	1.1%
51-100	3.0%	3.5%	10,682	13,217	2.0	2.0	1.3%
101+	18.8%	25.2%	12,943	16,859	2.4	2.6	1.6%
All workers	100.0%	100.0%	7,752	10,295	1.4	1.6	1.7%

Notes: Mean hourly income is expressed in constant 2025 pesos. Wage ratios divide each firm-size mean by the solo-worker mean in the same year. Annualized change is the compound annual change in mean hourly income between 2008 and 2025.

This shift toward the largest category is relevant beyond the wage-premium evidence. Recent growth theory emphasizes that the right tail of the firm-size distribution tends to thicken with economic development, and that large firms can matter for aggregate growth because they are central to idea search, diffusion, and productivity dynamics (Chen, 2026). In this light, Colombia’s increase in the employment share of 101+ firms is consistent with some movement toward a more developed firm-size structure. At the same time, the continued concentration of employment in solo work and micro-firms shows that this transformation remains incomplete.

Table 2 puts the raw firm-size gap in context by comparing it with remuneration gaps across sex, formality status, education, and economic activities in 2025. The wage gap by

sex is small in these weighted means: men earn 0.6 percent below the overall mean and women 0.8 percent above it. Other gaps are much larger. Formal workers earn 48.0 percent above the overall mean, while informal workers earn 39.1 percent below it; this contrast is close in magnitude to the 2.6-to-1 ratio between workers in 101+ firms and solo workers reported in Table 1. Education and sector differences are larger still. Workers with higher education earn 71.6 percent above the overall mean, while workers with preschool or less earn 57.9 percent below it. Across activities, agriculture is 41.4 percent below the mean, whereas information and communication is 112.8 percent above it. The raw firm-size gap is therefore far larger than the wage gap by sex, comparable to the formality gap, and smaller than the largest education and sector gaps.

Table 2: Mean real hourly labor income gaps by worker group, 2025

Dimension	Group	Employment share	Mean income	Gap	Ratio
Sex	Men	58.8%	10,236	-0.6%	1.00
	Women	41.2%	10,379	0.8%	1.01
Formality	Formal	44.9%	15,241	48.0%	2.43
	Informal	55.1%	6,271	-39.1%	1.00
Education	Preschool or less	2.9%	4,333	-57.9%	1.00
	Primary	18.1%	5,693	-44.7%	1.31
	Secondary	45.5%	7,085	-31.2%	1.63
	Higher education	33.5%	17,667	71.6%	4.08
Activity	A - Agriculture	13.8%	6,036	-41.4%	1.00
	T - Household employers	3.1%	6,454	-37.3%	1.07
	I - Accommodation and food	7.6%	6,930	-32.7%	1.15
	H - Transport	7.8%	7,775	-24.5%	1.29
	F - Construction	6.9%	8,406	-18.3%	1.39
	G - Commerce	17.0%	8,535	-17.1%	1.41
	R/S - Arts and other services	5.4%	8,765	-14.9%	1.45
	C - Manufacturing	11.0%	9,591	-6.8%	1.59
	D/E - Utilities and water	1.3%	11,295	9.7%	1.87
	B - Mining	1.3%	11,441	11.1%	1.90
	L/M/N - Business services	9.0%	13,459	30.7%	2.23
	Q - Health	4.4%	14,859	44.3%	2.46
	P - Education	4.2%	19,980	94.1%	3.31
	K - Finance and insurance	1.9%	21,389	107.8%	3.54
	O - Public administration	3.6%	21,508	108.9%	3.56
	J - Information and communication	1.7%	21,903	112.8%	3.63

Notes: Statistics use 2025 observations and GEIH expansion weights. Employment shares are computed within each dimension. Mean income is real hourly labor income in constant 2025 pesos. The gap is the percent difference in each group's weighted mean relative to the overall 2025 weighted mean. The ratio divides each group's mean by the lowest mean within the same dimension. Formality rows exclude pensioned occupied workers. Sector rows omit extraterritorial organizations.

Table 3 shows why the raw firm-size wage premium cannot be interpreted as a pure size effect. Larger firms employ a very different workforce. In 2025, only 8.3 percent of solo workers were formal, compared with 46.6 percent in firms with 6–10 workers and 98.0 percent in firms with 101 or more workers. Higher education also rises strongly with firm size, from 16.7 percent among solo workers to 62.9 percent in the 101+ category. The share of women varies less systematically across firm-size groups, while solo workers are older on average. These differences motivate the comparisons that follow, which ask whether the firm-size premium is visible within sex, formality, education, and sector groups.

Figure 1 looks at the firm-size wage premium by formality status. The bubble size

Table 3: Worker characteristics by firm size, 2025

Firm size	Women	Formal	Higher education	Mean age
Solo	41.3%	8.3%	16.7%	44.5
2-3	34.6%	13.0%	17.0%	41.3
4-5	36.5%	24.2%	20.2%	37.6
6-10	40.5%	46.6%	29.9%	37.0
11-19	40.4%	64.4%	38.8%	37.2
20-30	42.7%	81.2%	45.4%	37.0
31-50	41.2%	90.4%	51.0%	37.5
51-100	43.7%	94.3%	53.4%	37.5
101+	45.8%	98.0%	62.9%	38.0

Notes: Statistics use only 2025 observations and are weighted by GEIH expansion weights. Higher education corresponds to workers classified as superior or university educated.

represents the percentage of formal or informal workers in each firm-size category in the corresponding year. Formal workers earn more than informal workers and are much more concentrated in large firms. The wage pattern by firm size is still visible within informal workers: among them, mean hourly income rises from 5,677 pesos for solo workers to 12,813 pesos in firms with 101 or more workers in 2025. Among formal workers, the raw profile is closer to a U-shape: formal solo workers earned 15,990 pesos per hour in 2025, formal workers in firms with 4–5 workers earned 12,036 pesos, and formal workers in firms with 101 or more workers earned 16,944 pesos. Table 4 shows that formal solo work is a selected but heterogeneous category: 82.1 percent are own-account workers and 50.9 percent have higher education, with a sizable concentration in professional and administrative services, but 16.2 percent are domestic workers and 15.6 percent are in transport. Thus, the raw U-shape should be interpreted as evidence of worker composition within formal solo work, rather than as evidence that all formal solo workers are high-earning professionals.

Figure 2 shows that the raw firm-size pattern is similar for men and women. In 2025, among men, solo workers earned 6,816 pesos per hour, compared with 16,929 pesos in firms with 101 or more workers; among women, the corresponding means were 6,133 and 16,777 pesos. The large-firm advantage is therefore very similar for both sexes. The figure also shows that wage differences between men and women within a given firm-size category are small relative to the difference between solo work and large-firm employment.

Figure 3 shows that the raw firm-size premium is visible within every education group. Among workers with preschool or less, mean hourly income in 101+ firms is 2.1 times solo-worker income. The corresponding 101+-to-solo ratio is 1.8 among workers with primary education, 1.5 among workers with secondary education, and 1.7 among workers with higher education. Thus, the large-firm advantage is not only a comparison between more and less educated workers; it is also visible within each education level.

Figure 4 shows that the firm-size wage premium is widespread across economic activities. Mean hourly income in firms with 101 or more workers exceeded solo-worker income in 14 of 16 sectors in 2008 and in all 16 sectors in 2025. The two 2008 exceptions were finance and insurance and business services, where solo workers earned slightly more than workers in the largest firms; by 2025, the large-firm category also paid more in both sectors.

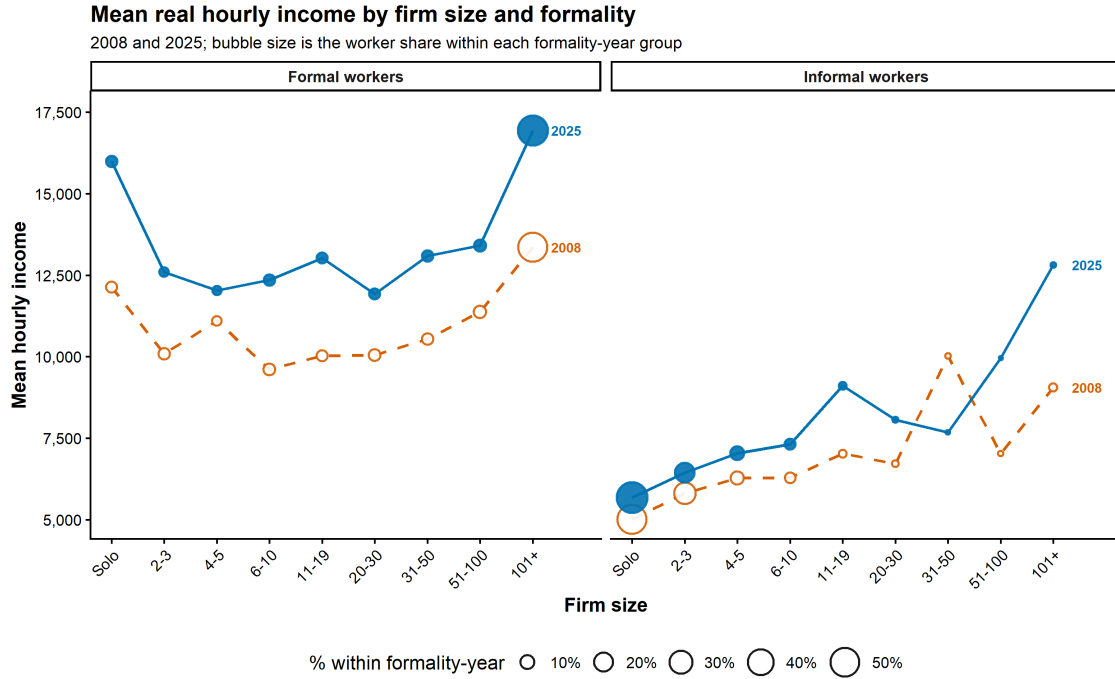


Figure 1: Mean real hourly income by firm size and formality status, 2008 and 2025

Table 4: Profile of formal workers by firm-size group, 2025

Characteristic	Formal solo	Formal 2–100	Formal 101+
Share of formal workers	6.4%	38.7%	54.9%
Mean real hourly income	15,990	12,696	16,944
Mean age	44.9	38.3	37.9
Women	43.3%	41.9%	45.7%
Higher education	50.9%	48.2%	63.4%
Own-account workers	82.1%	7.7%	9.6%
Domestic workers	16.2%	0.9%	0.0%
Professional and administrative services	22.8%	14.6%	9.4%
Transport and storage	15.6%	5.5%	5.6%
Commerce and repair	14.7%	21.1%	10.1%
Health and social assistance	5.2%	6.5%	10.6%
Households as employers	16.2%	0.9%	0.0%

Notes: Statistics use 2025 formal workers and GEIH expansion weights. The first row reports each column's share of all formal workers in the table. The 2–100 column pools all formal workers in firm-size categories from 2–3 through 51–100 workers. Higher education corresponds to workers classified as superior or university educated. Sector rows report the weighted share of each group located in the listed economic activity.

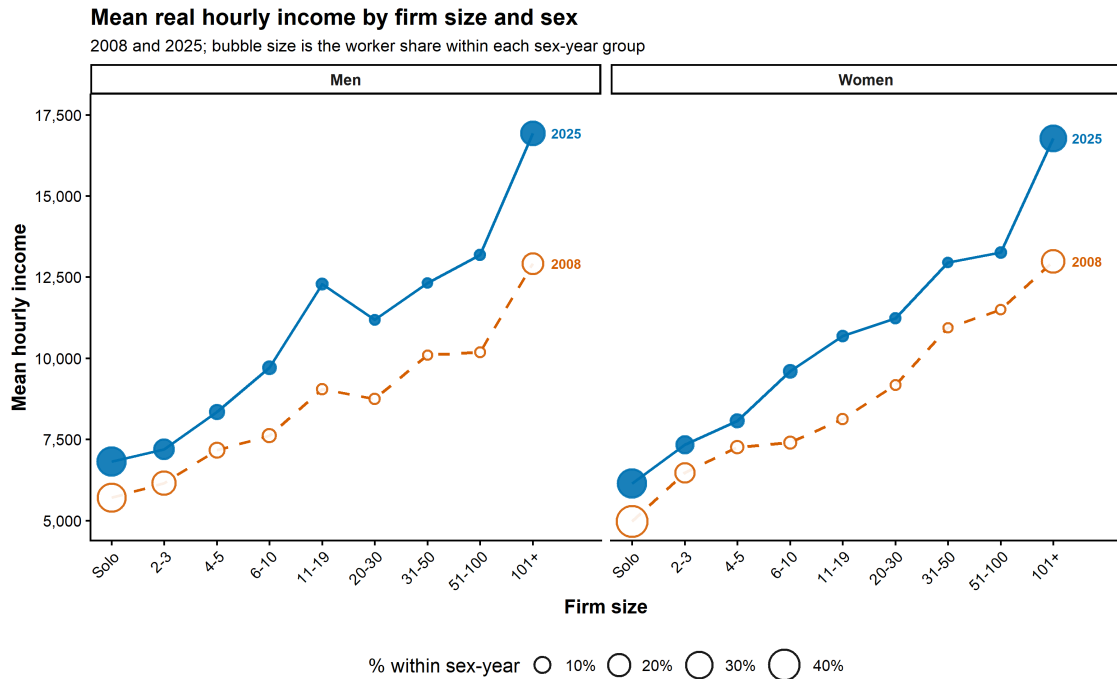


Figure 2: Mean real hourly income by firm size and sex

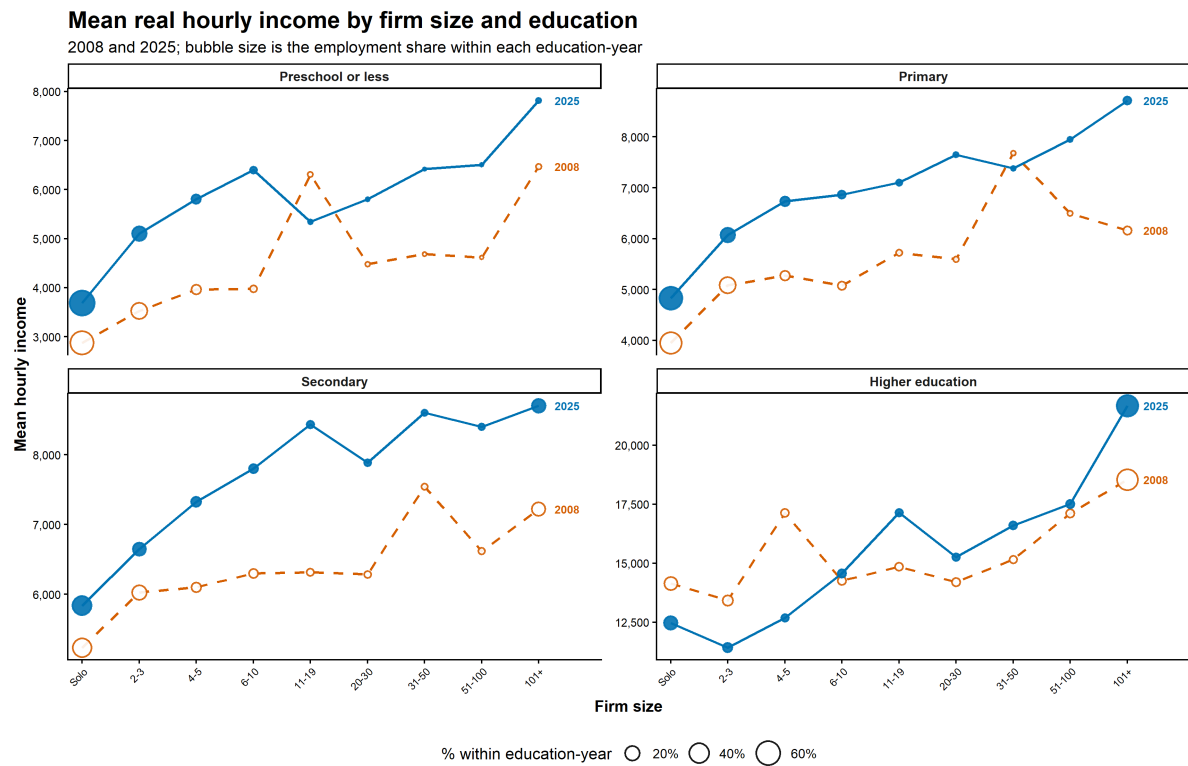


Figure 3: Mean real hourly income by firm size and education group

The size premium is especially pronounced in utilities and water, households as employers, mining, manufacturing, accommodation and food, and transport, while it is much flatter in information and communication, health, finance, and business services. These sector panels therefore reinforce the aggregate pattern while also showing that the size premium varies substantially across activities.

Taken together, the descriptive evidence motivates the regression analysis below. The central pattern is not only that mean hourly income is higher in larger firm-size categories. The tables and figures also provide suggestive evidence that a firm-size wage premium appears within each sex, within each education group, in most economic activities, and even within both formal and informal employment. At the same time, the descriptive results show strong sorting by formality, education, sector, and position in the firm-size distribution. The econometric exercise therefore asks how much of the raw firm-size premium survives after controlling for observable worker characteristics, formality, and sector-year conditions.

4 Methodology

4.1 Main Specification

The main econometric exercise estimates the conditional firm-size wage premium in a log-wage model. The baseline specification is:

$$\ln(w_{ist}) = \alpha + \sum_{g \neq g_0} \beta_g \mathbf{1}\{Size_{ist} = g\} + X'_{ist}\gamma + \lambda_{s \times t} + \varepsilon_{ist}, \quad (2)$$

where w_{ist} is real hourly labor income for worker i in sector s in year t , $Size_{ist}$ is the harmonized firm-size category, and g_0 is the reference group. We use solo workers as the reference group. This choice makes each coefficient directly interpretable as the conditional wage gap between workers in firm-size group g and workers at the bottom of the firm-size distribution.

The vector X_{ist} includes a female-worker indicator, age, age squared, education dummies, and a formal-employment indicator. The fixed effect $\lambda_{s \times t}$ absorbs sector-year conditions, allowing each broad sector to have its own wage level in each year. The model is estimated with GEIH expansion weights, and standard errors are clustered by sector.

The coefficients β_g are log-point differences relative to solo workers. For interpretation, we transform them into percentage premiums:

$$Premium_g = 100 \times \left(\exp(\hat{\beta}_g) - 1 \right). \quad (3)$$

4.2 Dynamic Premiums and Trend Test

The same framework can be extended to study the evolution of the premium over time by interacting firm-size categories with year indicators:

$$\ln(w_{ist}) = \alpha + \sum_{\tau} \sum_{g \neq g_0} \beta_{g\tau} \mathbf{1}\{Size_{ist} = g\} \mathbf{1}\{Year_t = \tau\} + X'_{ist}\gamma + \lambda_{s \times t} + \varepsilon_{ist}. \quad (4)$$

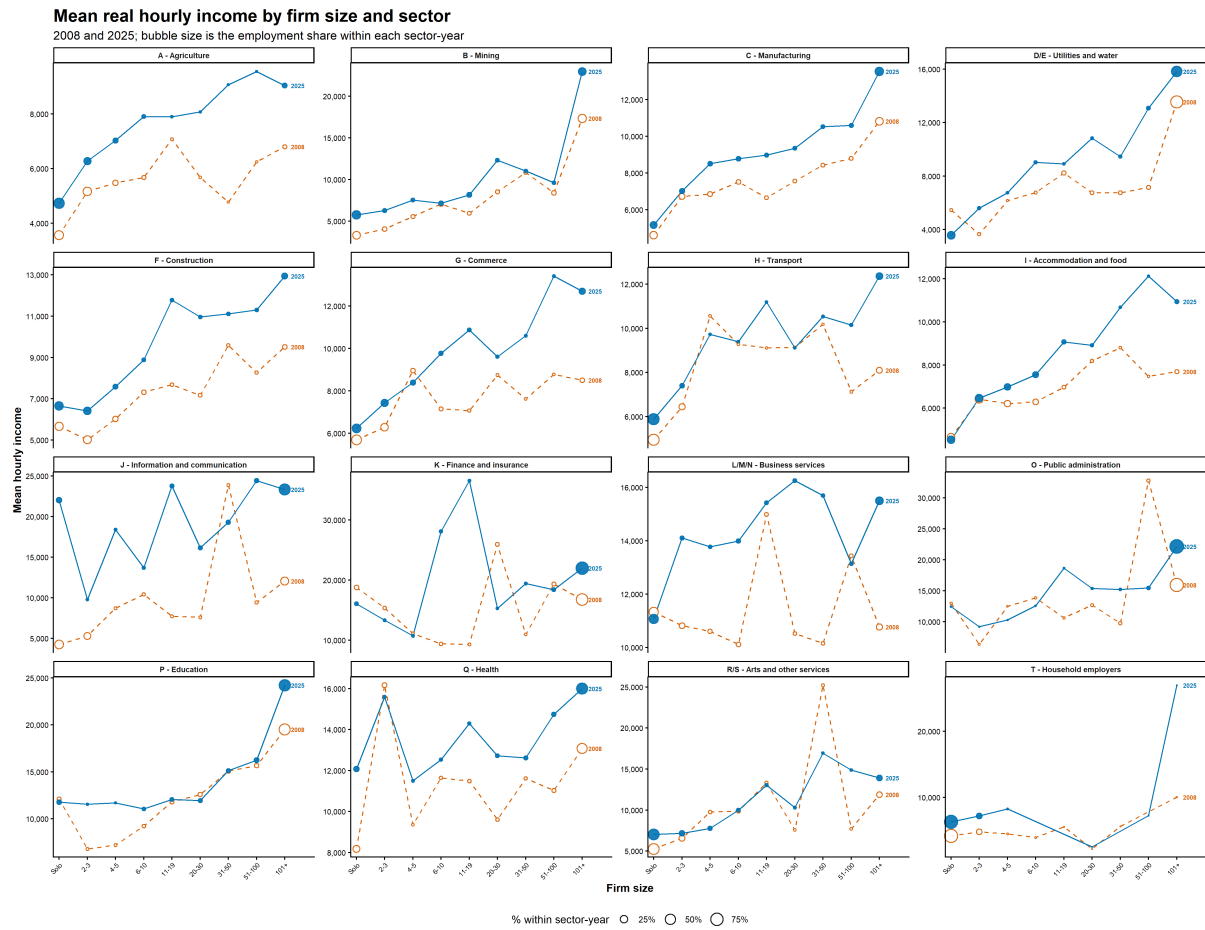


Figure 4: Mean real hourly income by firm size and sector, 2008 and 2025

This dynamic specification produces a sequence of firm-size premiums by year and can be used to assess whether the premium has widened, narrowed, or remained stable. We also estimate a more parsimonious trend specification:

$$\ln(w_{ist}) = \alpha + \sum_{g \neq g_0} (\beta_g + \delta_g Trend_t) \mathbf{1}\{Size_{ist} = g\} + X'_{ist}\gamma + \lambda_{s \times t} + \varepsilon_{ist}, \quad (5)$$

where $Trend_t$ is a linear year trend normalized to zero in 2008. The parameter δ_g measures whether the wage premium for firm-size group g changed over time relative to solo workers. Positive values of δ_g imply a steeper association between firm size and wages, while negative values imply a flatter one.

4.3 Firm-Size Nonlinearities by Formality Status

The descriptive evidence suggests that the relationship between firm size and hourly income may differ between formal and informal workers. We therefore estimate a specification that interacts all firm-size categories with formal employment:

$$\ln(w_{ist}) = \alpha + \phi Formal_{ist} + \sum_{g \neq g_0} (\beta_g + \theta_g Formal_{ist}) \mathbf{1}\{Size_{ist} = g\} + X'_{ist}\gamma + \lambda_{s \times t} + \varepsilon_{ist}. \quad (6)$$

The coefficients β_g measure the firm-size premium among informal workers, relative to informal solo workers. The corresponding premium among formal workers is $\beta_g + \theta_g$, relative to formal solo workers. The coefficient ϕ measures the formal-informal wage gap among solo workers, while $\phi + \theta_g$ measures the formal-informal wage gap within firm-size group g .

4.4 Approximate Firm-Size Elasticities

The main specifications treat firm size as a set of categories because this is how firm size is reported in GEIH and because the largest category is top-coded. This categorical approach is the most transparent way to estimate the Colombian firm-size wage premium. However, much of the employer-size wage effect literature reports a continuous elasticity of wages with respect to employer size. To compare the Colombian evidence with those benchmarks, we also estimate an approximate log-log specification:

$$\ln(w_{ist}) = \alpha + \eta \ln(\widetilde{Size}_{ist}) + X'_{ist}\gamma + \lambda_{s \times t} + \varepsilon_{ist}. \quad (7)$$

The variable $\widetilde{Size}_{ist}$ assigns representative values to the harmonized firm-size bins: 1 for solo workers, 2.5 for firms with 2–3 workers, 4.5 for 4–5 workers, 8 for 6–10 workers, 15 for 11–19 workers, 25 for 20–30 workers, 40.5 for 31–50 workers, 75.5 for 51–100 workers, and 150 for the top-coded 101+ category. The coefficient η is therefore an approximate elasticity: it measures the log-wage slope associated with assigned firm size. Because the underlying variable is grouped and top-coded, these elasticities are used only as benchmarking statistics. The categorical specifications remain the main empirical estimates.

5 Results

5.1 Main Results

Table 5 reports the regression results from the main specification in Equation (2). Column (1) shows the raw log-wage gap by firm size. The coefficient for firms with 101 or more workers is 1.019 log points, consistent with the large raw wage premium shown in the descriptive statistics. Column (2) adds sector-year fixed effects, reducing the coefficient for 101+ firms to 0.819 log points. Column (3) adds gender, age, age squared, and education controls, lowering the coefficient to 0.598. Column (4) adds formality, reducing the coefficient further to 0.359 log points, or 43.2 percent. Thus, roughly two thirds of the raw 101+ wage gap is absorbed by sector-year conditions and observed differences in worker and job characteristics, including gender, age, education, and formality. Nonetheless, a positive and statistically significant conditional premium remains after these controls.

Table 5: Firm-size wage premium regressions: the role of formality

	Dependent variable: log real hourly labor income			
	(1)	(2)	(3)	(4)
Firm size: 2-3	0.183*** (0.034)	0.242*** (0.030)	0.209*** (0.027)	0.188*** (0.029)
Firm size: 4-5	0.363*** (0.035)	0.408*** (0.030)	0.361*** (0.029)	0.307*** (0.038)
Firm size: 6-10	0.500*** (0.029)	0.502*** (0.033)	0.422*** (0.037)	0.317*** (0.050)
Firm size: 11-19	0.627*** (0.034)	0.577*** (0.040)	0.458*** (0.042)	0.303*** (0.051)
Firm size: 20-30	0.689*** (0.046)	0.604*** (0.045)	0.463*** (0.046)	0.271*** (0.050)
Firm size: 31-50	0.760*** (0.056)	0.651*** (0.049)	0.488*** (0.047)	0.272*** (0.048)
Firm size: 51-100	0.820*** (0.063)	0.699*** (0.052)	0.519*** (0.048)	0.289*** (0.047)
Firm size: 101+	1.019*** (0.102)	0.819*** (0.067)	0.598*** (0.061)	0.359*** (0.059)
Gender, age, and education controls	No	No	Yes	Yes
Formality control	No	No	No	Yes
Sector-year fixed effects	No	Yes	Yes	Yes
Observations	5,154,213	5,154,213	5,154,213	5,154,213
R^2	0.213	0.271	0.376	0.386

Notes: The omitted category is solo workers. All columns use the same estimation sample and GEIH expansion weights. Standard errors, clustered by sector, are reported in parentheses. Worker controls include a female-worker indicator, age, age squared, and education dummies. Column (4) adds labor formality. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

This attenuation pattern is consistent with the classic firm-size wage premium literature.

Brown and Medoff (1989) show that large employers pay more even after accounting for several observable worker and job characteristics, and Troske (1999) reaches a similar conclusion using matched worker-establishment data. Evidence from developing economies also points to sizable large-employer wage gaps. Schaffner (1998) documents premiums to employment in larger establishments in Peru, Söderbom et al. (2005) find that the earnings-firm size relationship remains substantial after accounting for individual fixed effects in Kenya and Ghana, and Reed and Tran (2019) show that the large-firm wage premium is widespread across developing economies. The Colombian context adds a further margin because firm size and formality are tightly connected. El Badaoui et al. (2010) show for Ecuador that what appears to be a formal-sector wage premium can be largely a firm-size wage differential, and Colombian firm-survey evidence emphasizes the role of firm characteristics in wage differentials within the formal labor market (Iregui et al., 2011). Our estimates fit this broader pattern. The fall in the coefficient after adding formality shows that the raw firm-size gap partly reflects segmentation between formal and informal jobs; the remaining premium shows that the Colombian firm-size wage gap is not only a formal-informal contrast.

Figure 5 transforms the coefficients of column (4) in Table 5 into percentage premiums. Relative to solo workers, the conditional premium is 20.7 percent for firms with 2–3 workers, 36.0 percent for firms with 4–5 workers, 37.3 percent for firms with 6–10 workers, roughly 31–35 percent for intermediate-size categories, and 43.2 percent for firms with 101 or more workers. All confidence intervals exclude zero.

Appendix Table 11 shows that this result is not driven by broad geographic differences across departments. The 101+ coefficient remains positive and statistically significant after adding department-year fixed effects, after absorbing sector-department-year fixed effects, and after adding occupational-position controls to that local-geography specification.

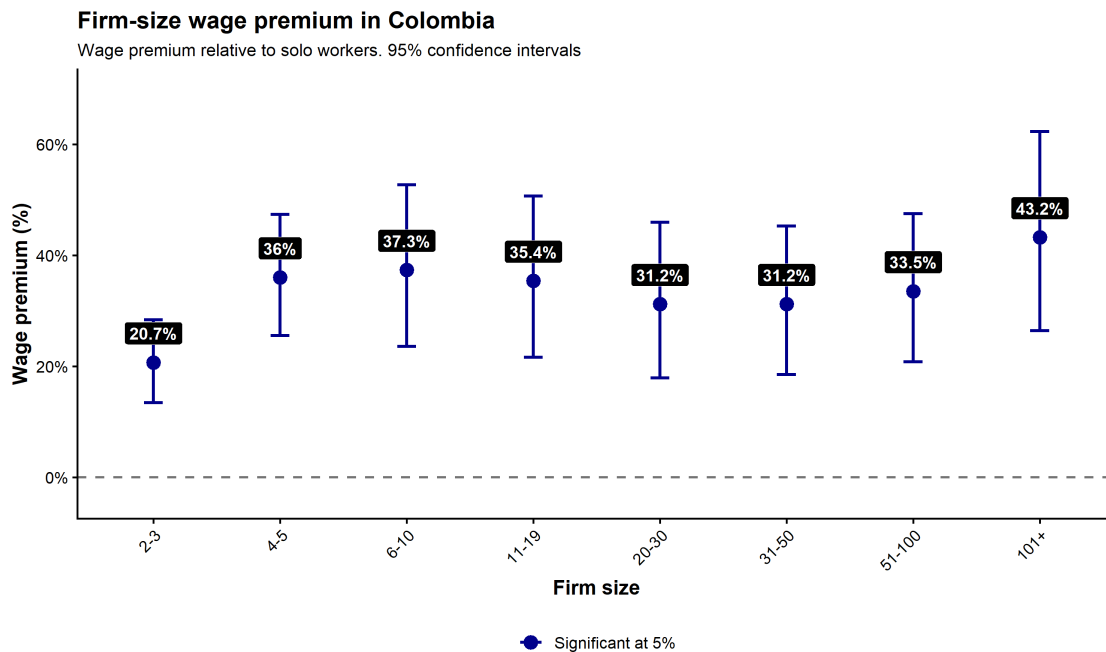


Figure 5: Conditional firm-size wage premium relative to solo workers

5.2 Dynamic Results

The dynamic exercise asks whether the firm-size wage gap steepened, flattened, or remained stable over time. A steepening premium means that firm size became more strongly associated with wages. A flattening premium means that the association weakened, although it does not identify whether the change came from faster wage growth in smaller firms or slower wage growth in larger firms.

Figure 6 estimates the full specification separately by interacting firm-size categories with year indicators. The conditional premium is positive throughout the sample period in all firm-size categories. The largest firms display the highest premiums in most years, although the confidence intervals are wider for these categories.

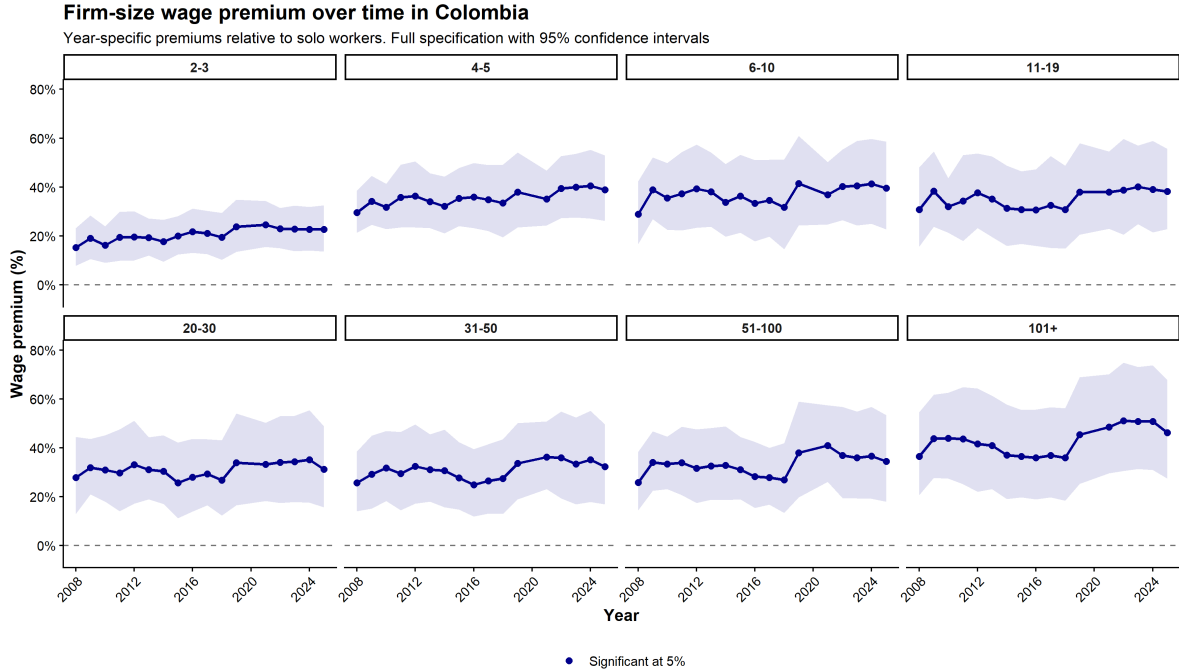


Figure 6: Conditional firm-size wage premium over time

Table 6 formally tests whether the premium changed over time using a linear trend interacted with firm-size categories. The estimated trend coefficients are positive for all size categories. The implied increase between 2008 and 2025 ranges from 3.2 percent for firms with 20–30 workers to 7.9 percent for firms with 101 or more workers. However, the two-sided tests do not reject a zero trend at the 5 percent level. The point estimates therefore suggest widening rather than narrowing, but they do not provide statistically conclusive evidence of a trend.

The evidence on the evolution of the large-firm wage premium over time is mixed. For the United States, Bloom et al. (2018) document a steady decline in the large-firm wage premium. Reed and Tran (2019) shows that large-firm wage premia are also common in developing economies, but tend to decline with national income and over time. Cross-country evidence points to heterogeneity rather than a single trajectory: Colonnelli et al. (2018) find a large-firm wage premium in Brazil, Germany, Sweden, and the United Kingdom, but no

common time path across countries. Evidence from Brazil also shows that firm-related wage differences can compress when institutions and labor-market conditions raise wages at the lower end of the distribution (Alvarez et al., 2018; Engbom and Moser, 2022). Against this background, Colombia looks less like a case of clear convergence or clear divergence and more like a case of persistence: firm scale remains strongly associated with hourly wages throughout the period, the point estimates suggest some widening, but the two-sided tests do not support the stronger claim that the premium systematically intensified.

Table 6: Testing whether the firm-size wage premium changed over time

Firm size	Annual trend	S.E.	p -value	Implied 2008–2025 change
2-3	0.0033	(0.0028)	0.262	5.7%
4-5	0.0032	(0.0031)	0.314	5.6%
6-10	0.0021	(0.0034)	0.536	3.7%
11-19	0.0027	(0.0039)	0.494	4.7%
20-30	0.0018	(0.0041)	0.661	3.2%
31-50	0.0027	(0.0044)	0.542	4.7%
51-100	0.0026	(0.0042)	0.549	4.5%
101+	0.0045	(0.0043)	0.314	7.9%

Notes: The annual trend is the coefficient on the interaction between firm-size category and a linear year trend, with solo workers as the omitted category. The specification controls for gender, age, age squared, education, formality, and sector-year fixed effects, and uses GEIH expansion weights. Standard errors are clustered by sector. The p -value corresponds to the two-sided test that the trend differs from zero. The final column reports $100 \times [\exp(17\hat{\delta}) - 1]$, the implied change in the firm-size wage ratio between 2008 and 2025.

5.3 Formality Nonlinearities

Table 7 reports the coefficients behind Equation (6). The estimates of β_g , which measure the firm-size premium among informal workers, are positive and statistically significant in every firm-size category. The estimates of θ_g are negative in every category, showing that the formal-worker profile is flatter than the informal-worker profile. The implied formal-worker premium $\beta_g + \theta_g$ is small and not statistically different from zero at the 5 percent level when formal solo workers are the reference category. This pattern shows a robust size premium among informal workers but a much flatter profile among formal workers.

Figure 7 translates these estimates into percentage premiums relative to solo workers within each formality group. Among informal workers, the conditional premium is positive and statistically significant in every firm-size category: it is 21.7 percent in firms with 2–3 workers, rises to about 48–51 percent across most categories from 6 to 100 workers, and remains 40.0 percent in firms with 101 or more workers. Among formal workers, the within-formality profile is much flatter and less precisely estimated: the estimated premium ranges from 2.1 to 7.0 percent for most categories and is 16.5 percent for firms with 101 or more workers, but the confidence intervals include zero throughout when each category is compared with formal solo workers.

This does not mean, however, that there is no evidence of conditional premiums within the formal sector. The U-shape in Figure 1 points to a different comparison: formal workers

Table 7: Firm-size coefficients by formality status

Firm size	$\hat{\beta}_g$	$\hat{\theta}_g$	$\hat{\beta}_g + \hat{\theta}_g$
2-3	0.197*** (0.026)	-0.176*** (0.062)	0.021 (0.067)
4-5	0.343*** (0.027)	-0.318*** (0.085)	0.024 (0.089)
6-10	0.392*** (0.032)	-0.369*** (0.081)	0.023 (0.087)
11-19	0.409*** (0.037)	-0.374*** (0.076)	0.035 (0.080)
20-30	0.402*** (0.035)	-0.379*** (0.072)	0.023 (0.079)
31-50	0.401*** (0.038)	-0.359*** (0.067)	0.041 (0.076)
51-100	0.410*** (0.044)	-0.342*** (0.067)	0.068 (0.075)
101+	0.336*** (0.046)	-0.183** (0.083)	0.153* (0.088)

Notes: Estimates correspond to Equation (6). The omitted category is solo workers within each formality group. $\hat{\beta}_g$ is the firm-size coefficient among informal workers. $\hat{\theta}_g$ is the additional coefficient for formal workers in the same firm-size category. $\hat{\beta}_g + \hat{\theta}_g$ is the implied firm-size coefficient among formal workers. All coefficients are in log points. Standard errors, clustered by sector, are reported in parentheses. The specification controls for gender, age, age squared, education, and sector-year fixed effects. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

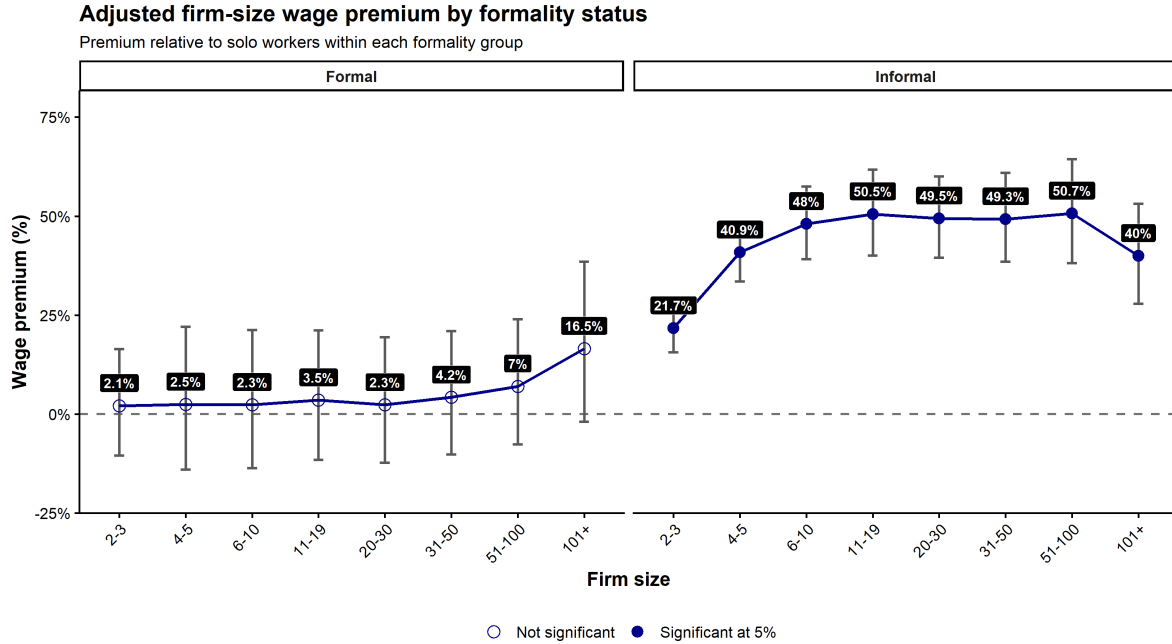


Figure 7: Conditional firm-size wage premium by formality status

in 101+ firms versus formal workers in intermediate-size firms. Direct contrasts from the interacted model show that formal workers in 101+ firms earn between 8.9 and 14.2 percent more than formal workers in every smaller non-solo category, and all of these non-solo contrasts are statistically significant, as reported in Table 8. Thus, the estimates show a large-firm premium relative to the middle of the formal firm-size distribution. The informal sector, by contrast, displays a robust conditional firm-size premium relative to solo work.

Table 8: Formal-worker large-firm wage premium relative to other formal firm-size categories

Reference category	Premium	<i>p</i> -value
Solo workers	16.5%	0.082
2-3	14.2%	< 0.001
4-5	13.8%	< 0.001
6-10	13.9%	< 0.001
11-19	12.6%	0.004
20-30	13.9%	0.001
31-50	11.8%	0.002
51-100	8.9%	0.004

Notes: Each row compares formal workers in firms with 101 or more workers with formal workers in the reference firm-size category. Estimates come from the firm-size-by-formality specification with gender, age, age squared, education, and sector-year fixed effects. Premiums are computed as $100 \times [\exp(\hat{\beta}) - 1]$ from the relevant linear contrast.

Table 9 reports the other side of the same specification: the conditional wage gap between formal and informal workers within each firm-size category. Formal workers earn more than informal workers in every category, and the differences are statistically significant throughout. The formal premium is 68.8 percent among solo workers, falls to roughly 16–23 percent through most intermediate firm-size categories, and rises again to 40.6 percent among workers in firms with 101 or more employees. Thus, the flatter formal-worker firm-size profile does not mean that formality is unimportant. Rather, formal employment shifts the wage level upward within each firm-size category, while the firm-size premium itself is much steeper among informal workers.

Figure 8 asks whether the adjusted formality-specific profiles changed between 2008 and 2025. The figure estimates the firm-size-by-formality specification separately for each end-point year, controlling for gender, age, age squared, education, and sector fixed effects, and plots the implied premium relative to solo workers within the same formality-year group. Among formal workers, the adjusted profile is flat in both years: in 2025, the estimates for all categories below 101+ range from -3.7 to 0.6 percent, and the 101+ estimate is 10.5 percent with a confidence interval that includes zero. Among informal workers, by contrast, the adjusted size profile is steep in both years and higher at the 2025 endpoint: in 2025, the premium rises from 24.3 percent in firms with 2–3 workers to 53.6 percent in firms with 101 or more workers. The figure therefore suggests that the strong conditional size premium within informal employment persists across the endpoints, while the formal-sector profile remains flat.

Figure 9 examines whether the pooled formal-worker result masks time variation. The figure plots the adjusted premium for firms with 101 or more workers relative to solo workers,

Table 9: Conditional formal-informal wage gap by firm size

Firm size	Log gap	Premium	<i>p</i> -value
Solo	0.524*** (0.063)	68.8%	< 0.001
2-3	0.348*** (0.018)	41.6%	< 0.001
4-5	0.205*** (0.036)	22.8%	< 0.001
6-10	0.155*** (0.031)	16.7%	< 0.001
11-19	0.149*** (0.024)	16.1%	< 0.001
20-30	0.145*** (0.018)	15.6%	< 0.001
31-50	0.165*** (0.014)	17.9%	< 0.001
51-100	0.181*** (0.019)	19.9%	< 0.001
101+	0.341*** (0.043)	40.6%	< 0.001

Notes: Each row reports the estimated wage gap between formal and informal workers within the same firm-size category from Equation (6). For solo workers, the gap is $\hat{\phi}$; for all other categories, it is $\hat{\phi} + \hat{\theta}_g$. Premiums are computed as $100 \times [\exp(\widehat{gap}) - 1]$. Standard errors, clustered by sector, are reported in parentheses below the log gap. The specification controls for gender, age, age squared, education, and sector-year fixed effects. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

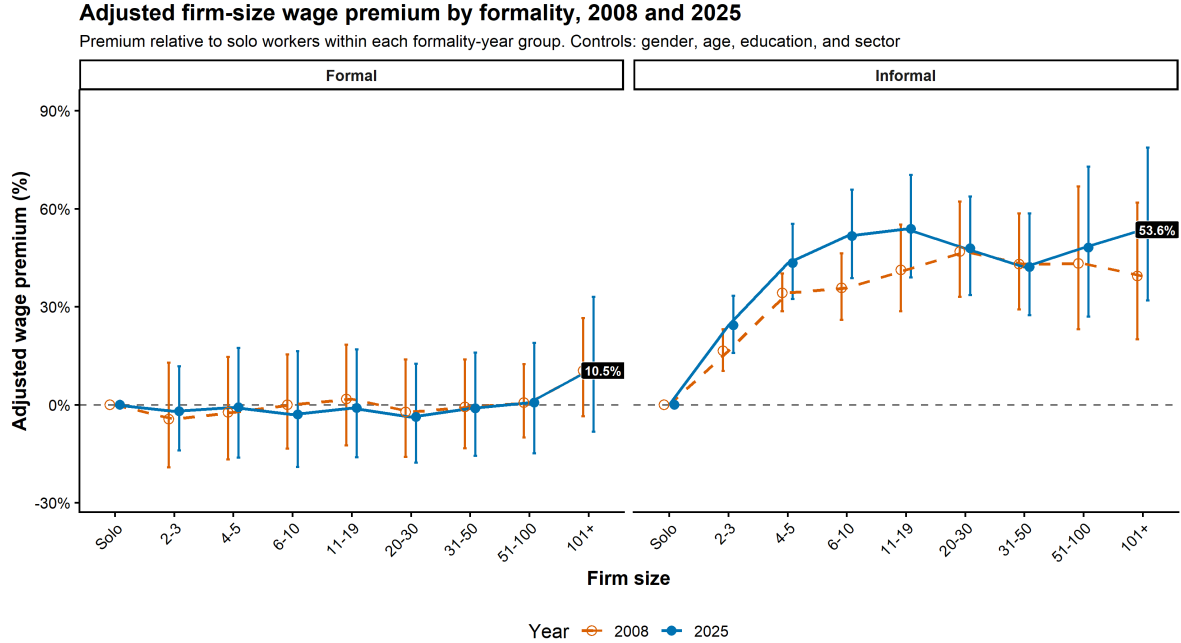


Figure 8: Adjusted firm-size wage premium by formality status, 2008 and 2025

separately within formal and informal employment in each year. The evidence does not show a reappearance of a statistically significant large-firm premium among formal workers in the most recent years. The formal 101+ premium is positive and significant in several earlier years, especially 2009–2013 and 2017, but after 2021 the point estimates are smaller and the confidence intervals include zero; in 2025, the estimate is 10.5 percent with a confidence interval from -8.2 to 33.0 percent. By contrast, the informal 101+ premium is positive and significant throughout the period, rising from 39.4 percent in 2008 to 53.6 percent in 2025.

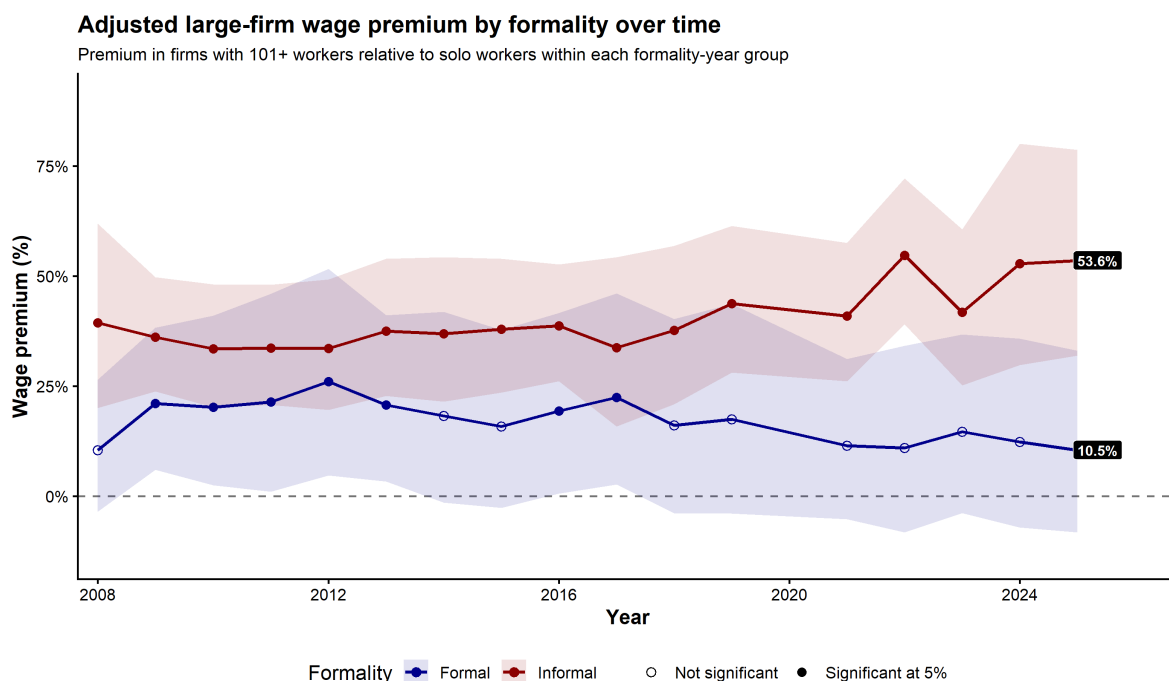


Figure 9: Adjusted large-firm wage premium by formality status over time; 2020 is omitted

This formal-informal contrast connects directly with the literature on formality and firm size. Formality can mediate the firm-size wage premium because larger firms are more visible to the state, more likely to comply with labor regulation, and more likely to offer jobs covered by social protection. Colombian evidence already points to persistent formal-informal wage gaps and to productivity differences associated with registration and regulatory compliance among microfirms (Daza and Gamboa, 2013; Gutierrez and Rodriguez-Lesmes, 2023). In that sense, part of the raw premium may reflect the fact that large-firm jobs are disproportionately formal jobs. From the complementary angle, El Badaoui et al. (2010) show for Ecuador that what appears to be a formal-sector wage premium can be largely explained by firm size. Formality can also moderate the firm-size premium: the wage gap by firm size need not be the same within formal and informal employment. Balkan and Tumen (2016) show for Turkey that the firm-size wage gap is more pronounced among informal jobs than among formal jobs. Our results are close to this second pattern. Adding formality in Table 5 reduces the overall 101+ coefficient substantially, which shows that formality mediates part of the aggregate size gap. Figures 7 and 8 show the complementary result: the conditional size premium remains especially strong among informal workers, while the formal-worker profile

is much flatter when each category is compared with formal solo workers.

5.4 Benchmarking the Firm-Size Wage Elasticity

Table 10 reports the approximate elasticity estimates from Equation (7) and compares them with the two most directly comparable benchmarks from the meta-analysis of Diegmann et al. (2026): specifications without worker-heterogeneity controls and specifications with observable worker controls. The comparison should be read as a magnitude check rather than as a direct meta-analytic test: the Colombian estimates are based on worker-reported firm-size bins, while the studies summarized by Diegmann et al. (2026) estimate continuous employer-size elasticities with different data structures and controls.

Table 10: Benchmarking Colombia’s approximate firm-size wage elasticity

Source	Specification	Elasticity	S.E.
Diegmann et al. (2026)	No controls	0.062	n.a.
	Worker controls	0.036	n.a.
Colombia (GEIH)	No controls	0.203***	(0.019)
	Worker controls - Full sample	0.071***	(0.013)
	Worker controls - Formal workers	0.026**	(0.011)
	Worker controls - Informal workers	0.130***	(0.015)

Notes: Diegmann et al. (2026) benchmarks correspond to the mean elasticities reported in their Table 1 for specifications without worker-heterogeneity controls and with observable worker controls. Colombian elasticities are approximate log-log slopes estimated with GEIH expansion weights. The full-sample Colombian specification includes sector-year fixed effects, worker controls, and formality; the formal- and informal-worker specifications include sector-year fixed effects and worker controls. Because GEIH reports firm size in bins, the estimates assign representative values of 1, 2.5, 4.5, 8, 15, 25, 40.5, 75.5, and 150 workers to the harmonized categories from solo work to 101+ workers. Standard errors for the Colombian estimates are clustered by sector. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The Colombian raw binned elasticity is 0.203, far above the mean no-controls elasticity of 0.062 in the literature summarized by Diegmann et al. (2026). After sector-year effects, worker controls, and formality are added, the full-sample Colombian elasticity falls to 0.071, but remains above their observable-controls benchmark of 0.036. The formality split helps interpret why the Colombian full-sample elasticity is high relative to that benchmark. The difference is not driven equally by formal and informal employment. Among formal workers, the estimated elasticity is 0.026, close to the benchmark. Among informal workers, it is 0.130, more than three times the benchmark. Thus, the comparison with the meta-analysis clarifies that Colombia’s high conditional elasticity is mainly an informal-sector phenomenon.

6 Discussion

The results link firm size to three related outcomes: wage inequality, labor-market segmentation, and possible allocative frictions. The inequality result is direct: workers in larger

firm-size categories earn more than workers in solo work and micro-firms, both in raw averages and after conditioning on observed characteristics. The segmentation result comes from the role of formality. Adding formality to the main regression absorbs part of the raw large-firm premium, and formal workers earn more than informal workers within every firm-size category. At the same time, the premium does not disappear after controlling for formality, and the size profile differs sharply between formal and informal employment.

The formality results narrow the policy-relevant interpretation. Formal employment raises wage levels within every firm-size category, but the conditional size premium is steepest among informal workers. Within formal employment, the premium relative to formal solo workers is small and not statistically significant, although workers in 101+ firms earn more than workers in intermediate-size formal firms. If the conditional wage gaps partly reflect productivity-related frictions rather than only unobserved sorting, the evidence points to three margins for possible productivity gains: moving workers and jobs from informality to formality, scaling informal micro-units into larger and more organized firms, and, more selectively, growth or movement within the formal sector from intermediate-size firms toward larger employers.

These interpretations must remain cautious, though. A wage premium that survives observable controls does not identify a single mechanism as an explanation. As Brown and Medoff (1989) and Oi and Idson (1999) emphasize, the remaining gap may reflect unobserved worker quality, firm productivity, rent sharing, efficiency wages, or match quality. The developing-country evidence cautions against reducing the premium to sorting alone: Söderbom et al. (2005) find that the firm-size effect remains substantial in Kenya and Ghana after controlling for individual fixed effects. Matched employer-employee studies also show that employer-associated characteristics matter for wage variation (Abowd et al., 1999; Card et al., 2013; Song et al., 2019), and related evidence from Brazil reaches a similar conclusion for Latin America (Alvarez et al., 2018; Gerard et al., 2021; Engbom and Moser, 2022). Recent German evidence sharpens this caution: Diegmann et al. (2026) find that the employer-size wage effect disappears when establishment size is compared within the same multi-establishment firm, suggesting that the premium may reflect employer-wide wage policies, rents, or other firm attributes rather than establishment size per se. The elasticity benchmark in Section 5.4 is consistent with a large Colombian size-wage association, especially among informal workers, but GEIH does not identify employers or follow workers across firms. The paper therefore cannot separate worker sorting from other competing mechanisms, so the estimated premiums and elasticities should be interpreted as diagnostic magnitudes rather than causal effects of firm size.

The evidence is also not a direct test of misallocation. Misallocation in the strict sense requires comparing marginal products across firms or sectors, as in Hsieh and Klenow (2009). The link with allocative efficiency comes from the marginal-product condition: in an efficient allocation, labor should move across firms until the marginal product of labor is equalized across uses. In the competitive benchmark, wages reflect the value of marginal product, so persistent wage differences across firm sizes are informative about possible allocative frictions. If larger firms pay more than smaller firms for workers with the same observable characteristics, this may indicate that labor is more valuable in larger firms or that barriers prevent workers from moving toward higher-productivity matches. The evidence is only a marker, not proof, because wages may also reflect unobserved worker quality or other

unobserved features of the employment relationship. Testing misallocation directly would require firm-level data on output, capital, and employment, which would allow us to estimate marginal products rather than infer them indirectly from wages.

The policy implication is therefore not that policy should favor large firms as such or push workers mechanically into large firms. The relevant objective is to reduce barriers that keep productive firms small, informal, or disconnected from larger markets, while also improving workers' access to better firm matches. The formality results sharpen this point. Because formal employment raises wage levels within every firm-size category, moving jobs from informality to formality remains a policy priority. Because the firm-size premium is steepest among informal workers, a second policy priority is helping productive informal micro-units become larger, more organized, and more connected to markets. This support should be designed as a path toward formalization, not as a subsidy for remaining informal; otherwise, it could weaken incentives for formal firms to remain formal. The evidence gives less support to a policy focus on scaling up formal firms: within formal employment, the premium relative to formal solo workers is small, although workers in 101+ firms earn more than workers in intermediate-size formal firms. This pattern is consistent with a segmented labor market in which the strongest wage-based signs of allocative frictions are concentrated in informal employment. The largest efficiency gains therefore appear to lie in formalization and in the growth of productive informal firms, rather than in firm growth within the formal sector.

7 Conclusion

This paper documents a clear firm-size wage premium in Colombia using harmonized GEIH microdata for 2008–2019 and 2021–2025. The descriptive evidence shows that real hourly wages rise sharply with firm size, while employment remains concentrated in solo work and micro-firms. Solo workers and workers in firms with up to ten workers account for about two thirds of employment, and the middle of the firm-size distribution remains thin.

The regression results show that a substantial conditional premium remains after controlling for gender, age, age squared, education, formality, and sector-year fixed effects. In the full specification, workers in firms with 101 or more employees earn about 43 percent more than otherwise comparable solo workers. The premium is positive throughout the sample period. The point estimates suggest an increase between 2008 and 2025, but the estimated change is not statistically significant.

The formal-informal results sharpen the main finding. Formal workers earn more than informal workers within every firm-size category. At the same time, the firm-size premium is much steeper among informal workers than among formal workers. Within formal employment, the premium relative to formal solo workers is small, although workers in 101+ firms earn more than workers in intermediate-size formal firms. Taken together, these results show that the Colombian firm-size wage premium is not only a formal-informal contrast. It is a persistent wage gap associated with firm scale, especially within informal employment, and it remains after conditioning on the main observable worker and job characteristics available in the GEIH.

A Additional Regression Table

This appendix reports the robustness checks referenced in Section 5. Table 11 first adds department-level fixed effects. In Colombia, departments are first-level territorial units, with Bogotá treated as a department-equivalent Capital District. The most demanding geography specification absorbs sector-department-year fixed effects, so the firm-size coefficients are identified within the same sector, department, and year. The final column adds occupational-position controls to this specification. The large-firm coefficient remains positive and statistically significant: workers in firms with 101 or more employees earn about 40.2 percent more than comparable solo workers after conditioning on sector-department-year cells and occupational position.

Table 11: Firm-size wage premium robustness checks

	Dependent variable: log real hourly labor income			
	(1)	(2)	(3)	(4)
Firm size: 2-3	0.188*** (0.011)	0.168*** (0.011)	0.167*** (0.011)	0.102*** (0.013)
Firm size: 4-5	0.307*** (0.014)	0.277*** (0.014)	0.273*** (0.014)	0.226*** (0.015)
Firm size: 6-10	0.316*** (0.018)	0.282*** (0.017)	0.278*** (0.017)	0.265*** (0.016)
Firm size: 11-19	0.302*** (0.018)	0.272*** (0.017)	0.267*** (0.017)	0.272*** (0.015)
Firm size: 20-30	0.271*** (0.019)	0.241*** (0.017)	0.237*** (0.017)	0.258*** (0.015)
Firm size: 31-50	0.271*** (0.018)	0.245*** (0.017)	0.240*** (0.017)	0.264*** (0.015)
Firm size: 51-100	0.289*** (0.018)	0.262*** (0.017)	0.257*** (0.017)	0.282*** (0.015)
Firm size: 101+	0.359*** (0.023)	0.338*** (0.024)	0.325*** (0.024)	0.338*** (0.022)
Worker and formality controls	Yes	Yes	Yes	Yes
Occupational-position controls	No	No	No	Yes
Fixed effects	$s \times t$	$s \times t, d \times t$	$s \times d \times t$	$s \times d \times t$
Observations	5,268,514	5,268,514	5,268,375	5,268,375
R^2	0.386	0.404	0.414	0.429

Notes: The omitted category is solo workers. The sample is restricted to observations with nonmissing department codes. Worker controls include a female-worker indicator, age, age squared, education dummies, and labor formality. The local geography d is the department, treating Bogotá as a department-equivalent Capital District. Column (3) absorbs sector-department-year fixed effects, so identification comes from wage differences across firm-size categories within the same sector, department, and year. Column (4) adds occupational-position controls to this most stringent local-geography specification. Standard errors are clustered by sector-department cells. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Data and Code Availability

Replication code and materials are available at:

<https://github.com/Kin-George/CJC-Monitor/tree/working-paper-v1-2026-06-13>.

Raw GEIH microdata are not redistributed and should be obtained directly from DANE.

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